

Abdullah Al Mahmud Joy

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[LinkedIn](#) — [Google Scholar](#) — [ORCID](#) — [GitHub](#)

PROFESSIONAL SUMMARY

Computer Science graduate and M.Sc. student in Computer Science and Engineering at the Military Institute of Science and Technology (MIST). Experienced in software testing, API testing, artificial intelligence research, and academic activities. Author of four IEEE peer-reviewed publications in Explainable Artificial Intelligence (XAI), Machine Learning, and Computer Vision. Hands-on experience with Postman, Jira, Python, TensorFlow, and software engineering methodologies. Passionate about software engineering, research, and continuous learning.

EDUCATION

Military Institute of Science and Technology (MIST) Ongoing

M.Sc. in Computer Science & Engineering

Bangladesh University of Business and Technology (BUBT) Jan 2022 – Dec 2025

B.Sc. in Computer Science & Engineering

CGPA: 3.92/4.00

Birshreshtha Munshi Abdur Rouf Public College Jul 2017 – Jun 2019

Higher Secondary Certificate (HSC)

GPA: 5.00/5.00

Adarsha High School, Mirpur Jan 2015 – May 2017

Secondary School Certificate (SSC)

GPA: 5.00/5.00

EXPERIENCE

Teaching Assistant Jul 2025 – Dec 2025

Bangladesh University of Business and Technology

- Assisted faculty members during programming and software engineering laboratory sessions.
- Guided undergraduate students in solving programming and technical problems.
- Maintained academic records and documentation with accuracy.
- Collaborated with faculty members to support teaching and learning activities.
- Strengthened communication, teamwork, and mentoring skills.

Independent Researcher 2024 – Present

- Conduct research in Explainable Artificial Intelligence (XAI), Machine Learning, and Computer Vision.
- Develop and evaluate deep learning models for medical imaging and agricultural applications.
- Prepare IEEE conference manuscripts and collaborate on peer-reviewed research publications.
- Perform literature reviews, experimental evaluation, and comparative model analysis.

PROJECTS

DMoney Manual Testing GitHub

- Designed and executed manual test cases for digital wallet functionalities.
- Performed functional and exploratory testing on transaction workflows.
- Reported defects with severity, priority, reproduction steps, and supporting evidence.
- Prepared test scenarios, test cases, bug reports, and testing summary documentation.

DMoney API Testing with Postman & Newman GitHub

- Designed and executed API test collections using Postman.
- Validated authentication, transaction APIs, status codes, and response bodies.
- Automated API workflows using environment variables and assertions.
- Generated Newman reports and documented testing results.

Alibaba E-Commerce Website Manual Testing

GitHub

- Designed and executed manual test cases for major user workflows
- Identified defects and prepared detailed bug reports with reproduction steps
- Prepared test scenarios and testing documentation following QA best practices

Steadfast Courier Mobile App Testing

GitHub

- Performed manual testing of core mobile application features
- Conducted exploratory testing to identify functional and usability issues
- Documented defects and maintained structured testing reports

RESEARCH PUBLICATIONS

1. **Joy, A. A. M.**, M. F. Islam, and M. M. Hossain. "Ensemble Deep Learning for Brain Tumor Classification Using MRI: A Comparative Study of CNN Architectures." *IEEE COMPAS 2025*. *IEEE Xplore*
2. **Joy, A. A. M.**, M. F. Islam, M. S. Islam, S. Ahmed, M. N. Morshed, and H. Ahmed. "CropNet-XAI: An Explainable 1D-CNN Framework for Transparent Crop Yield Prediction." *IEEE ICECTE 2026*. *IEEE Xplore*
3. **Joy, A. A. M.**, M. F. Islam, A. Rahman, M. S. Islam, and M. M. Hossain. "XAI-Crop: Explainable Counterfactual Crop Recommendation via Ensemble Learning." *IEEE ICECTE 2026*. *IEEE Xplore*
4. **Joy, A. A. M.**, M. F. Islam, and M. S. Islam. "Detecting Potholes with Convolutional Neural Networks and Transfer Learning." *IEEE ICECTE 2026*. *IEEE Xplore*

TECHNICAL SKILLS

Programming	Python, C, SQL
Software Testing	Manual Testing, API Testing, Postman, Newman, Jira, Basic Selenium, Basic TestNG
Machine Learning	TensorFlow, Keras, Scikit-learn, OpenCV
Software Engineering	SDLC, STLC, Functional Testing, Regression Testing, Exploratory Testing, Bug Reporting
Research Tools	LaTeX, IEEE Xplore, Google Scholar, Microsoft Office
Version Control	Git, GitHub

RESEARCH INTERESTS

Explainable Artificial Intelligence (XAI), Machine Learning, Deep Learning, Computer Vision, Software Quality Assurance, Medical Image Analysis, Agricultural AI, Predictive Analytics

ACHIEVEMENTS

- Participated in ICPC Asia Dhaka Regional Contest 2023.
- Participated in BUBT Intra Department Programming Contest in 2022 and 2023.
- Solved over 300 problems across various online judges.
- Awarded tuition fee waiver scholarships 8 times for academic excellence.